

Feedback Loops

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Handle no match for Implementing a recommender. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A recommender shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

6. Solve the given case

A recommender shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the feedback loop.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 24 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A recommender shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace selection

Why does A win the worked example even though B is processed first?

My reasoning and evidence

13. W2 Handle no match

State `recommend([], 10)` and the worked items with limit 8.

My reasoning and evidence

14. W3 Repair order dependence

A tied selection changes after reversing input order. Identify the likely missing policy and test.

My reasoning and evidence

15. W4 Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

My reasoning and evidence

16. W5 Explain the result

Write a one-sentence explanation of X winning without calling it objectively best.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Feedback Loops

A fictional catalogue repeatedly recommends two books using observed clicks.

Initially each click count is 0. Equal counts choose earlier ID A over B.

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.

Exposure and clicks after the invented three rounds

	Shown	Clicked
A	3	3
B	0	0

Read a cell using both its row and column labels.

Feedback Loops

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial tie

Counts $A=0, B=0$; ID rule shows A.



2. Record event

A receives a click, so its count becomes 1; B stays 0.



3. Repeat selection

Higher A count causes A to be shown again; after three clicks $A=3, B=0$.



4. Interpret carefully

B's zero may mean no exposure, not dislike.

CHANGE ONE THING

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

What extra field would reveal why comparing raw clicks is misleading?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Log recommender exposure, choice and explicit feedback separately; consider whether its own policy narrows future evidence.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Log recommender exposure, choice and explicit feedback separately; consider whether its own policy narrows future evidence.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue repeatedly recommends two books using observed clicks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's zero may mean no exposure, not dislike.

1. Record event
2. Repeat selection
3. Initial tie
4. Interpret carefully

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Interpret carefully B's zero may mean no exposure, not dislike. The selection policy changes which evidence is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books. Keep display counts separate from click counts; investigate alternatives with a teacher-approved exposure plan rather than treating zero as negative feedback.

Practice 2: explain the deciding evidence

Changed-case practice: On a planned trial, show B once and record no click. Which conclusion is supported by the supplied facts?

1. B gains one exposure but still zero clicks.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: B gains one exposure but still zero clicks. The observation now supports a nonclick on that display, not a permanent dislike.

Repair a missing building block

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing.. Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
